

A brain-wide genome-wide association study of candidate quantitative trait loci associated with structural and functional phenotypes of pain sensitivity

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Individual pain sensitivity is modulated by the brain's structural and functional features, but its heritability remains unclear. This paper conducted a brain-wide genome-wide association study (GWAS) to explore the genetic bases of neuroimage phenotypes of pain sensitivity. In total, 432 normal participants were divided into high and low pain sensitivity groups according to the laser quantitative test threshold. Then, the brain's gray matter density (GMD) features correlated with pain sensitivity were identified. Next, GWAS was performed on each GMD phenotype using quality-controlled genotypes. Based on the heatmap and hierarchical clustering results, the right insula was identified for further refined analysis in terms of subregions GMD and resting-state functional connectivity (rs-FC) phenotypes. The results indicate that the right insula GMD in the high sensitivity group is significantly lower than that in the low sensitivity group. Also, the TT/TC group at locus rs187974 has lower right insula GMD than the CC group. Further, loci at gene CYP2D6 may lead to a variation of rs-FC between the right insula and left putamen. In conclusion, our study suggests that the right insula and multiple candidate loci may be importantly involved in pain sensitivity modulation, which may guide the future development of precision pain therapeutics.

Key words: pain sensitivity; imaging genetics; GWAS; gray matter density; functional connectivity.

Introduction

Pain is highly variable across individuals, and it is determined by many factors, such as the differentiation of regional features of the brain, genetic variation among the population, different psychiatric conditions, and even personalities (Rainville 2002; Coghill 2010; Zhang et al. 2021). Among these factors, the brain's structure and function as well as genetic variation have been well documented to have large effects on pain sensitivity. On the one hand, the variation of brain structural and functional features may alter the way of pain perception since the brain processes the pain sensation. On the other, many single-nucleotide polymorphisms (SNPs) have been found to have substantial impacts on pain sensitivity (Foulkes and Wood 2008). Therefore, to better understand individual differences in pain sensitivity, it is important to study the underlying brain imaging and genetic basis of pain sensitivity and to further provide mechanistic explanations of these differences.

Many brain imaging studies have shown that pain perception is largely dependent on individuals' sensitivity to pain, and differences in pain sensitivity are often manifested by corresponding differences in the brain (Nielsen et al. 2009; Hu and Iannetti 2016). Based on various types of brain imaging techniques, such as electroencephalography (EEG) and magnetic resonance imaging (MRI), former studies have successfully linked individual pain sensitivity to the variability in brain structure and function. Structural MRI studies have indicated that morphologic features of

the brain, such as gray matter (GM) density (GMD) and cortical thickness, are associated with pain sensitivity in healthy subjects (Erpelding et al. 2012; Emerson et al. 2014; Zhang et al. 2021). Further, some fMRI studies indicated that regional blood oxygen level-dependent (BOLD) signal variability is inversely correlated with individual pain sensitivity within the ascending nociceptive pathway (primary and secondary somatosensory cortices), default mode network, and salience network (Rogachov et al. 2016). In addition, other research had established the heritable nature of both chronic pain and experimentally induced pain (Kato et al. 2006). In the meantime, several pain studies have shown an association between individual genetic variations and variability in pain perception. For example, twin studies have suggested that the heritability of different pain traits is about 50%, which implies that the variability of pain sensitivity is likely related to some specific genes (MacGregor et al. 2004). Though, in recent years, many genes have been proposed to have associations with different types of pain, most studies conducted the analysis based on small groups of candidate genes (Young et al. 2012). A relatively large-scale imaging genetics database on pain sensitivity is still lacking. Consequently, further studies that unite brain features and genetic data on identifying pain sensitivity-related brain regions and genes are still very limited.

Imaging genetics for pain and other respects of brain-related research is an emerging interdisciplinary research field, which has shed light on how genetic variation influences brain structure and

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functions (Shen and Thompson 2019). One of the great advantages of using brain imaging genetics methods based on genome-wide association study (GWAS) is to combine brain imaging and genomics data with other phenotype data so as to reveal risk loci and genes that may be related to diseases and behavior traits (Visscher et al. 2017; Nakai 2019). GWAS makes use of high-throughput genotyping technology to relate a large number of SNPs to brain imaging quantitative traits (QTs) (Uffelmann et al. 2021), which can be more efficient than traditional case-control studies. To date, only a few imaging genetics works were conducted for pain-related research (Johnston et al. 2019; Fontanillas et al. 2022), which impeded the exploration of neural mechanisms associated with genetic variation by fusing genes with imaging information. Since pain perception is processed in the brain (Zou et al. 2021), it is of significance to reveal the genetic and neuroimaging mechanisms of pain sensitivity as well as to investigate their underlying relationships, such as GMD and function connectivity (FC) from different pain sensitivity groups (Glahn et al. 2007; Ziv et al. 2010). As a powerful technique, GWAS is able to associate genetic variation with imaging features to investigate the genetic determinants of pain sensitivity for different brain structural and functional phenotypes.

Therefore, in this work, we focus on imaging genetics study using a self-collected dataset and mainly address the following 3 issues on pain sensitivity: First, identify brain regions that are significantly associated with laser pain sensitivity; second, identify loci and genes that are related to the imaging QTs from brain structural and functional modalities; third, reveal the associations and effects among the genes, QTs, and pain sensitivity. Specifically, inspired by previous research (Shen et al. 2010), we present a brain-wide GWAS using the laser quantitative test threshold to explore the brain and genetic architecture of pain sensitivity. GWAS was carried out on 432 normal participants, which were divided into high and low pain sensitivity groups according to their laser pain threshold from the laser quantitative test. Accordingly, brain structural and resting-state fMRI (rs-fMRI) images were acquired from an MRI scanner. A separate GWAS analysis using PLINK was completed for each of these phenotypes. Afterward, a heatmap and hierarchical clustering were used to demonstrate the significant associations between SNPs and phenotypes at threshold $P \leq 10^{-6}$. Subsequent detailed analysis showed that several brain regions are associated with potential key loci from the heatmap results. Since many previous studies reported that the insula plays a key role in pain perception (Ostrowsky et al. 2002), such a region and its subregions were selected for further analysis (Deen et al. 2011). A series of refined analyses using two-way ANOVA were performed to analyze the relationship between key loci, brain phenotype (GMD and FC), and pain sensitivity. Our results suggest that the brain-wide genome-wide association approach can reveal novel candidate genes and loci that may be related to pain sensitivity. The results show that individuals with a potential risk allele and high pain sensitivity at a key locus may have different structural and functional features in the corresponding brain regions. Overall, our work can help to understand the mechanism and physiology of pain sensitivity more comprehensively from the perspective of both imaging features and genetics.

Materials and methods

Participants

Figure 1 shows the overall flowchart of the data collection and analysis methods of our study. We recruited 432 healthy

participants (259 females, age: 21.30 ± 3.99 years) via college and community advertisements for the experiment. Twenty left-handed participants were excluded due to differences in brain morphology and function among left-handed and right-handed participants (Pool et al. 2015; Margiotoudi et al. 2019; Wiberg et al. 2019). Before the experiments, participants were carefully screened to ensure that they had no history of chronic pain, neurological diseases, cerebrovascular diseases, coronary heart disease, or mental disorders, and they had no contraindications to MRI examination. The study was proved by the local ethics committee and all participants gave their written informed consent before participating in the study. The data are not publicly available as the participants of this study did not give written consent for their data to be shared publicly.

Extraction and genotyping

Five-milliliter blood samples were collected from participants for DNA extraction and sequencing. Genotypes of SNPs were determined with the CapitalBio Technology Corporation Precision Medicine Research Array based on the Affymetrix platform. After the above processing, 748,659 SNP markers were collected as genome-wide genotype information.

Pain sensitivity definition

The pain sensitivity of all the participants was measured as laser pain thresholds using quantitative sensory testing before the MRI scan. A series of infrared neodymium:yttrium aluminum perovskite (Nd:YAP) laser stimuli were delivered to the back area between the thumb and index finger of a participant's left hand. The laser pain threshold was acquired using quantitative sensory testing. The measurement was started from an energy level of 1 J with a 0.25 J increase at each stimulus. After each stimulus, a participant was asked to report the pain rating from 0 (no pain) to 10 (unbearable pain). When a rating of 4 was reported, the corresponding energy level was recorded as the laser pain threshold of the participant. For each participant, the laser pain threshold was averaged from 2 independent measurements conducted in 1 h.

The values of the laser pain threshold were normalized by z-score normalization to determine the pain sensitivity threshold to meet the normal distribution. Based on the above threshold, the participants were first sorted from the low pain threshold (high sensitivity to pain) to the high pain threshold (low sensitivity to pain) and were then divided into 2 groups based on the 0 threshold.

MRI data acquisition

MRI data were obtained on a 3.0-Telsa system (GE Medical Systems, Milwaukee, WI, United States) at the Brain and Cognitive Neuroscience Research Center, Liaoning Normal University, Dalian, China. A standard birdcage with an 8-channel head coil, along with restraining foam pads, was used to minimize head motion and scanner noise. High-resolution structural T1-weighted images were collected using a 3D magnetization-prepared rapid gradient echo sequence with the following imaging parameters: flip angle = 8° , field of view = 256×256 mm², data matrix = 256×256 , in-plane resolution = 1×1 mm², slices = 176, and slice thickness = 1 mm. Ten-minute resting-state functional images were acquired using an echo-planar imaging (EPI) sequence with the following imaging parameters: repetition time = 2,000 ms, echo time = 29 ms, flip angle = 90° , field of view = 192×192 mm², data matrix = 64×64 , in-plane resolution = 3×3 mm², and slice thickness = 3 mm. During the rs-fMRI data acquisition, participants have presented with a screen with a

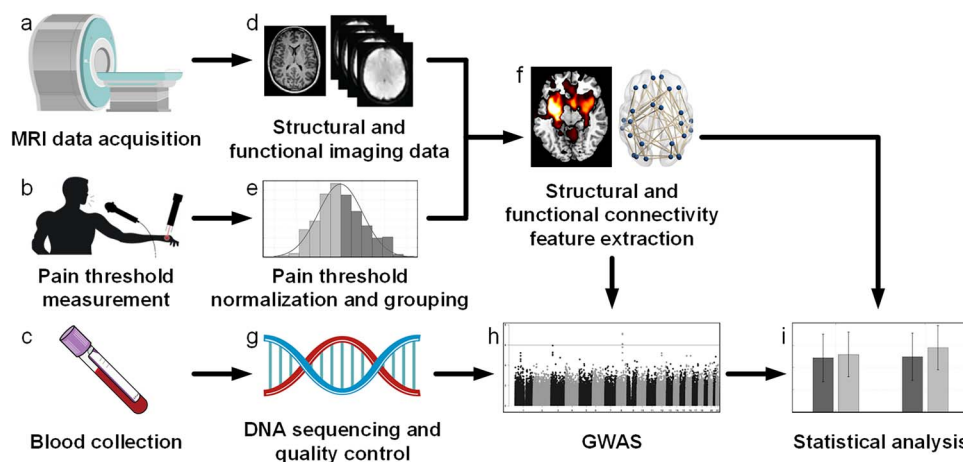


Fig. 1. The overall flowchart of our data collection and analysis method; a–c) collecting MRI data, pain threshold and blood sample of each subject; d) obtaining structural and functional imaging data after preprocessing procedures; e) grouping the subjects into high and low pain sensitivity groups based on normalized pain threshold; f) extracting pain-related structural and functional connectivity phenotype features based on MRI data and normalized pain threshold; g) DNA sequencing and genetic data QC; h) GWAS analysis for structural and functional connectivity phenotypes; i) statistical analysis for identified genotype–phenotype associations.

white crosshair in the center of the black background. They were instructed to relax, remain awake, and not think about anything with their eyes open.

Processing of structural MRI data

The raw DICOM scans were converted into the Neuroimaging Informatics Technology Initiative format using MRICRON software after review. We performed the preprocessing steps using CAT12 toolboxes under SPM12 (Statistical Parametric Mapping, <http://www.fil.ion.ucl.ac.uk/spm>) with the default setting. Voxel-based morphometry (VBM) (Good et al. 2001) was used to process and extract brain-wide target MRI imaging phenotypes. The structural images were segmented into GM, white matter (WM), and cerebrospinal fluid (CSF) components by applying a registration to the Montreal Neurological Institute (MNI) stereotactic space and subsequent nonlinear deformation. The nonlinear deformation parameters were calculated via the inbuilt high-dimensional Diffeomorphic Anatomic Registration Through Exponentiated Lie (DARTEL) Algebra algorithm (Ashburner 2007). Then, the warping functions generated by DARTEL were used to spatially normalize the GM segments and we modulated them by the Jacobian determinant. Finally, the normalized GM images were smoothed using an 8-mm full width at half maximum Gaussian kernel. After the above processing, the total GM volume, WM volume, and CSF volume were obtained from the smoothed images in the CAT12 toolboxes and were saved in the total intracranial volume (TIV).

A regression analysis was performed using a general linear model to examine the relationship between GMD across the whole brain and pain sensitivity (Emerson et al. 2014). Age, gender, and TIV were regressed out as covariates. Then, brain regions with a significant relationship between GMD and normalized pain threshold ($P < 0.05$) were extracted as relevant regions of interest (ROIs) for subsequent association analysis.

Processing of fMRI data

The rs-fMRI data were preprocessed with data processing & analysis for brain imaging (<http://www.rfmri.org/dpabi>) (Yan et al. 2016). First, the initial 10 volumes of the time series were removed due to the stability of the gradient magnetic field and participants' adaptation to the scanner circumstances. The remaining images were slice-timing corrected, realigned, and resliced to correct for

head motion. Participants were excluded if their head motion was >2.0 mm of translation or 2.0° of rotation throughout scanning for further analysis (16 participants were excluded). Then, all fMRI images were spatially normalized into the MNI space using the EPI template, and each voxel was resampled to isotropic $3 \times 3 \times 3$ mm³. After normalization, all datasets were smoothed with a Gaussian kernel of $8 \times 8 \times 8$ FWHM. Next, the following nuisance covariates from each voxel's time course were regressed out, including 24 head motion parameters (Friston et al. 1996), WM, and CSF signals as well as the linear trend. Subsequent data required temporal filtering in a frequency range of 0.01 to 0.1 Hz to reduce the effects of low-frequency drift and high-frequency noise.

Brain parcellation and extraction of multimodality imaging phenotypes

The automated anatomical labeling (AAL) atlas 3 (Rolls et al. 2020) was used to create the ROI-level measurements. For structural MRI data, the GMD at the level of 170 ROIs was extracted from GMD images as structural image phenotypes. For rs-fMRI data, BOLD time series of whole-brain were obtained for resting-state functional connectivity (rs-FC) phenotypes. More specifically, for each participant, the time series was extracted as the mean value across all voxels within that ROI. For each ROI, the Pearson correlated coefficient (r -value) between itself and all other ROIs in the AAL3 template was calculated by the voxel-voxel method (Allen et al. 2014). Then, we converted the r -values to z -values by applying Fisher's r -to- z transformation (Liu et al. 2019). Finally, the z -values between pairs of ROIs were used as the rs-FC phenotypes.

Quality control

To handle possible defects of genetic data, the following quality control (QC) steps were performed on these genotype data using the PLINK software package (v1.9) (<https://zzz.bwh.harvard.edu/plink/>) (Purcell et al. 2007). First, only SNPs located at autosomes were included in this study and 27,932 SNPs at sex chromosomes were removed. Second, SNPs were excluded if they could not meet any of the following criteria: (i) call rate per SNP $\geq 90\%$ (46,803 SNPs removed); (ii) minor allele frequency $\geq 5\%$ (319,751 SNPs removed); and (3) Hardy–Weinberg equilibrium test of $P \leq 10^{-6}$ (752 SNPs removed). Participants were excluded from the

analysis if any of the following criteria was not satisfied: (i) call rate per participant $\geq 90\%$ and (ii) gender check (5 participants were excluded). Population stratification analysis (Freedman et al. 2004) was performed by applying the principal component analysis (PCA), which is considered to be a very useful method of adjusting spurious associations. This method enables us to obtain the ethnic information of the sample and to determine possible ethnic outliers. After the QC procedure, 407 out of 412 participants and 353,421 out of 748,659 SNP markers remained in the analysis.

Genome-wide association analysis

The GWAS analysis was conducted separately on structural and rs-FC phenotypes after the QC procedures of the SNP data. Before that, the phenotypes were adjusted for the baseline age, gender, TIV or head motion, and PCA parameters. We applied the PLINK software package to calculate the main effects of all SNPs on the corresponding quantitative imaging phenotype with the QT association option. An additive SNP effect was assumed, and the empirical P-values were based on the Wald statistics (Purcell et al. 2007).

To investigate the GWAS results and data reduction for subsequent refined analyses, we employed a heatmap to visualize the association pairs between identified SNPs and corresponding imaging phenotypes at different statistical significance levels. A color map was used to represent various significance levels. The association between an SNP and an imaging phenotype can be regarded as significant if $P \leq 10^{-6}$ is found (Shen et al. 2010). Based on the obtained heatmap result, a hierarchical clustering method was applied to identify the similarity among SNPs and imaging phenotypes, respectively. The clustering was completed using the Euclidean distance method to define the dissimilarity between 2 nodes. The distance average among all pairs of objects in 2 clusters was used to measure the similarity between the 2 clusters. In this way, 2 dendrograms can be obtained to show the clustering results for imaging phenotypes and SNPs, respectively. All association results surviving the significance threshold of $P \leq 10^{-6}$ were saved and were prepared for additional pattern analysis.

Following many previous pain-related studies, which reported that the insula plays a key role in the emotional-affective dimension of pain and for pain modulation (Ostrowsky et al. 2002), the GMD of the right insula was selected for further refined analysis. It is because the number of significant SNP associated with the right insula was the highest. A Manhattan plot and a quantile-quantile (Q-Q) plot were used to visualize the GWAS results for the right insula. Moreover, since former works show that each subregion of the insula has different cell structure and functional distributions in pain perception (Mazzola et al. 2012; Wiech et al. 2014; Lu et al. 2016), bilateral insulae were further divided into 3 subregions in subsequent analyses through data-driven clustering techniques (Deen et al. 2011). The above method was performed in both structural and resting-state functional image features. Based on Menon et al. (2020), we obtained 3 insula subregions in the right hemisphere, including the posterior insula (PI), the dorsal anterior insula (dAI), and the ventral anterior insula (vAI), by merging the related regions for further statistical analysis.

Refined analysis

After extracting the genotype information, refined analyses were further conducted to quantitatively determine the difference of multimodality phenotypes between various genotype groups. More specifically, the participants were divided into 3 groups based on different loci, including 2 homozygous groups and

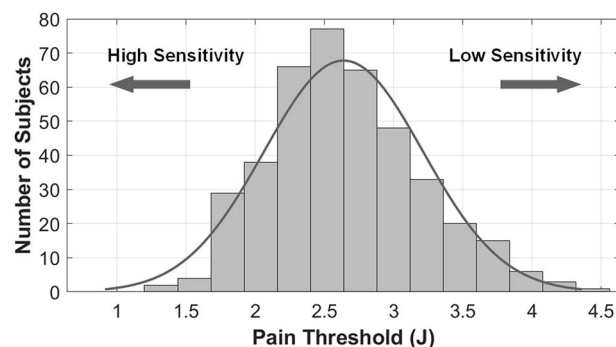


Fig. 2. Histogram with a normal distribution fit of pain threshold of 407 participants. The curve is the fitting result of the data with a Gaussian distribution.

1 heterozygous group. Due to the uneven distribution of the number of participants in different groups, the homozygous group with a small number of participants was combined with the heterozygous group as a risk allele-carrier group (Liu et al. 2019). After grouping participants by genotype, a two-sample t-test was used to determine the association between genotype and phenotype characteristics. Groups of high and low pain sensitivity were then divided, and a two-sample t-test was used to examine the association between pain sensitivity and phenotypic features. Finally, a two-way ANOVA (2×2 ANOVA: pain sensitivity $\times$ genotypes) was performed to examine the effect of target SNP and pain sensitivity on both GMD and rs-FC phenotype of target ROI. All statistical analyses were performed using SPSS 22.0 software.

Results

Sample characteristics after QC

After QC procedures of the genotyping data, 407 out of 432 recruited participants remained in the present study. Figure 2 shows the distribution and the fitting result using a Gaussian distribution of the pain threshold of the 407 participants. The pain thresholds are sorted from low (high pain sensitivity) to high (low pain sensitivity). The figure demonstrates that the number of participants with high pain sensitivity is relatively greater than that with low pain sensitivity. Then, the pain thresholds were normalized by applying z-score transformation. Interested readers may refer to Section 1 of the [Supplementary Document](#) for normalized pain threshold results. The structural and fMRI data from all these 407 participants were preprocessed, and the preprocessed GMD and rs-FC values were used as phenotypes for subsequent GWAS analysis. Table 1 illustrates the demographic information of the sample analyzed for both structural and resting-state functional studies. Among the participants, the genders of the participants are significantly different ($P = 0.027$) between the high and low sensitivity groups after the independent-sample t-test when dealing with structural phenotypes. In the subsequent GWAS analyses, age, gender, TIV, or head motion were considered as the covariates.

Extraction of ROIs associated with pain

Multiple regression of normalized pain threshold and GMD in unrestricted ROIs in the whole brain was performed based on SPM12 (adding age, gender, and head motion as covariates). Figure 3 shows that the identified brain regions were basically symmetrical. Meanwhile, regions of pain matrix have been

Table 1. Demographic information and the total number of participants are included in each analysis. After QC, GMD was extracted as a structural phenotype from the images of 407 participants, and rs-FC values were extracted as a functional phenotype from the resting-state MRI of 390 participants. Basic demographic information is shown for both groups of subjects.

Category	Structural phenotypes (407 participants)			rs-fMRI phenotypes (390 participants)		
	High	Low	P-value	High	Low	P-value
Number of participants	216	191	–	210	180	–
Gender (M/F) ^a	74/142	86/105	0.027 ^b	73/137	79/101	0.067
Age (years, mean $\pm$ SD) ^c	21.1 $\pm$ 3.9	21.4 $\pm$ 4.1	0.508	21.2 $\pm$ 3.9	21.2 $\pm$ 3.6	0.869
Education (year, mean $\pm$ SD) ^c	14.6 $\pm$ 2.0	14.5 $\pm$ 1.9	0.746	14.6 $\pm$ 2.1	14.6 $\pm$ 2.0	0.765

^aChi-square test. ^bP-value < 0.05. ^cTwo-sample t-test.

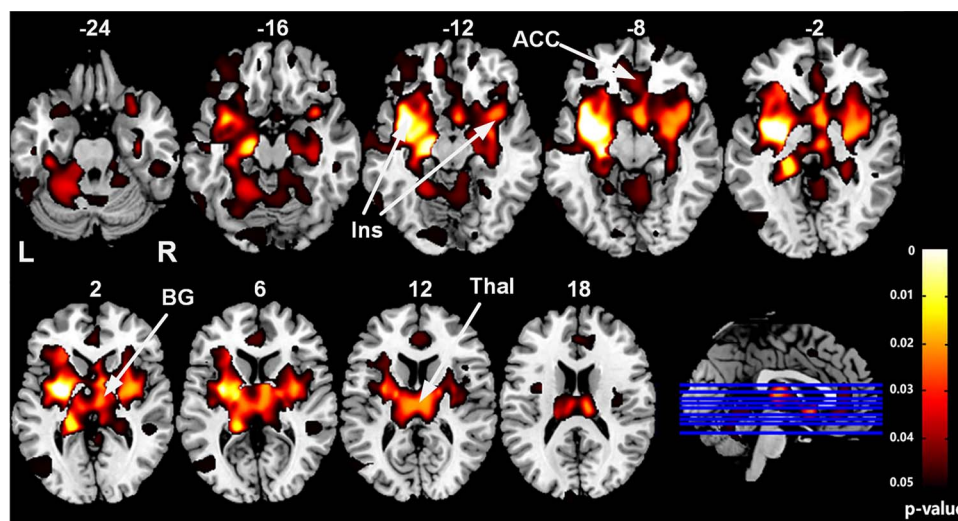


Fig. 3. Relationship between normalized pain threshold and GMD after multiple regression. Normalized pain threshold is positively correlated with GMD, while pain sensitivity is inversely related to GMD. Several brain regions belonging to the pain matrix have been indicated, including insulae (Ins), ACC, basal ganglia (BG), and thalamus (Thal).

indicated in Fig. 3 (Reddan and Wager 2018). Moreover, both left and right insulae correlated with normalized pain threshold significantly ($P < 0.05$, uncorrected). Then, GMD of 98 brain regions was extracted as structural phenotypes for subsequent GWAS analysis.

GWAS of GMD

The heatmap shown in Fig. 4 illustrates the imaging genetics associations at a significance threshold of $P \leq 10^{-6}$, which were discovered by GWAS analysis of 98 structural phenotypes, i.e. brain imaging QTs. The names of identified brain regions and SNPs are shown accordingly. On the heatmap, significant associations between imaging phenotypes and SNPs are marked with an “x” if $P \leq 10^{-6}$. At this significance level, 50 strong SNP-QT associations (blocks labeled with “x”) were identified in the GWAS analyses, and 46 SNPs were involved in these associations. Two dendrograms for the hierarchical clustering results of imaging phenotypes and SNPs can be obtained along with the x-axis and y-axis, respectively. The color bar on the left side of the heatmap encodes the chromosomes for the corresponding SNPs.

From Fig. 4, we can see that several imaging phenotypes were identified to have strong associations with target SNPs in the GWAS analyses. From the imaging dimension, bilateral insulae and amygdala regions, which are considered to be related to pain perception and modulation can be identified. Also, it is interesting to find that a group of regions such as caudate, pallidum, and supplementary motor area can also be identified. It suggests

that these regions are QTs associated with individual pain sensitivity. Observed from the genomic dimension, several SNPs can be identified from the results, including rs10504722, rs16907751, rs396760, and rs187974. It can be seen from the heatmap that all these 4 SNPs are associated with the right insula.

The hierarchical clustering results shown in Fig. 4 indicate that many pairs of left and right measures of the same region were clustered together as shown in the imaging dimension (x-axis). This suggests the symmetric relationship between these phenotypes and genetic variations. Moreover, several SNP groups from the genomic dimension (y-axis) in the same color can be identified from chromosome 8 (rs10504722, rs16907751, rs396760, and rs187974). This is probably caused by the linkage disequilibrium (LD) effect that increases the likelihood of nonrandom association of different alleles.

The right insula was selected as a target QT for further detailed analysis because it has the largest number of SNPs and a strong association with these SNPs. In Fig. 5a, a Manhattan plot of the GWAS for the target QT is shown. The x-axis of the Manhattan plot denotes different chromosomes, and the y-axis denotes the P-value of all SNPs. The Manhattan plot indicates that several potential SNPs located at chromosome 8 may be associated with pain sensitivity. Correspondingly, the Q-Q plot of the right insula is shown in Fig. 5b. From the figure, we can see that, for most of the P-values, the observed P-values from GWAS are almost the same as expected from the null hypothesis. The P-values in the upper tail of the distribution show a significant deviation. It suggests

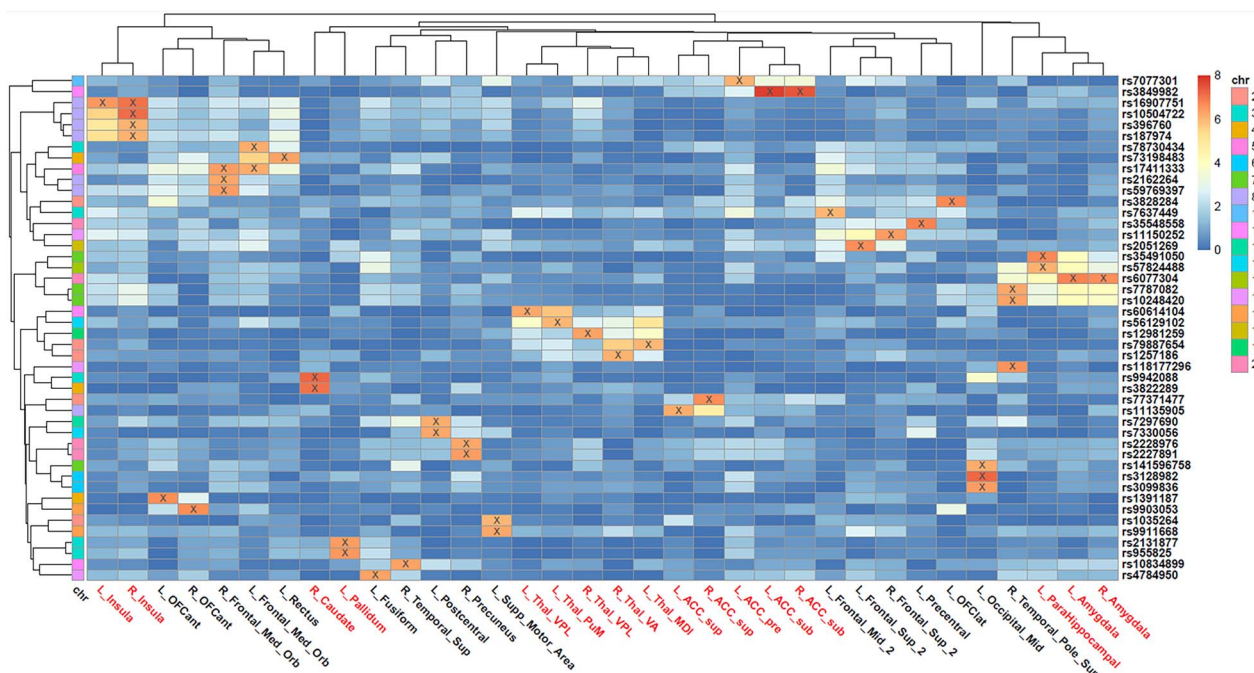


Fig. 4. Heatmap of SNP associations with QTs at the $P \leq 10^{-6}$ significance threshold. The blocks with “x” indicate the significance level of $P \leq 10^{-6}$. Only significant SNPs and QTs are shown in the heatmap. Dendrograms from the x-axis and y-axis were derived from hierarchical clustering for QTs and SNPs, respectively. The color bars on the right side encode the P-values and corresponding chromosomes for SNPs, respectively. The brain region names in red are often considered to be part of the pain matrix.

strong associations between these SNPs and the GMD of the right insula. Figure 5c shows the relationship between pain sensitivity and GMD in the right insula, and we can see that participants with higher normalized pain thresholds were found to have larger GMD in the target QT. It means that in the right insula, GMD and pain sensitivity are negatively correlated. The insula was further divided into 3 subregions in the right hemisphere to identify the morphological abnormalities. The GMD of each insula subregion was extracted and was used to correlate with genotypes separately.

Refined analysis for SNP with structural phenotype

We included 4 target SNPs (rs10504722, rs16907751, rs396760, and rs187974), which correspond to the right insula from the heatmap, for further imaging analysis. Only the target SNP rs187974 shows main effect for both genotype groups and pain sensitivity groups. As shown in Fig. 6, a two-way ANOVA using SPSS was performed to analyze the effect of the pain sensitivity group and rs187974 genotype on GMD of the right insula ($n=407$; 202 high pain sensitivity (39 TX, 173 CC); 183 low pain sensitivity (36 TX, 147 CC); 2 missings). In the figure, the mean and standard deviation (SD) of GMD are shown. Since the number of participants with a TT genotype in rs187974 is too small, participants with TT and TC genotypes were combined into the TX group. A main effect of genotype shows that the GMD of the CC group (as shown in light gray) is elevated more than that of the group of T-allele carriers (as shown in dark gray) of the right insula ($P = 0.0105$). Moreover, the participants with high pain sensitivity have lower GMD than those with low pain sensitivity ($P = 0.0487$). Accordingly, it may be suggested that if a participant is high-pain sensitive and carries the potential high-risk allele T at rs187974, the GMD of the subject may be lower than others.

After performing the GWAS analysis on GMD of subregions of the right insula and SNP data, only 4 SNPs (rs10504722, rs16907751, rs396760, and rs187974) were found to be significantly associated with their corresponding dAI. These SNPs were consistent with those obtained when the entire insula was analyzed, suggesting that the main effect of GMD and SNP association is on the dAI. As shown in Fig. 7, the homozygous group with a small number of participants is also integrated with the heterozygous group. The results illustrate that the participants with high pain sensitivity carrying the T-allele in the rs16907751 and rs187974 SNPs ($P = 0.0002$ and $P = 0.0079$, respectively) may have lower GMD in dAI than others. In addition, paired comparisons showed that the GMD of the dAI was lower in participants with risk allele A than in participants with G-allele, and the GMD of subjects with higher sensitivity was lower than those with lower pain sensitivity. This indicates that participants with high pain sensitivity carrying A-allele in rs396760 SNP ($P = 0.0101$) may have lower GMD in the dAI than others. Meanwhile, the result shows that participants with high pain sensitivity carrying risk C-allele in rs10504722 ($P = 0.0037$) have lower GMD of dAI than others.

Refined analysis for SNP with functional phenotype

The subjects' rs-FC of pain-related brain regions are highly correlated with individual pain sensitivity (Icenhour et al. 2017). Since the insula is a functionally diverse brain region that plays a central role in multiple aspects of pain processing (Segerdahl et al. 2015), it is crucial to investigate the rs-FC variations among subjects with different pain sensitivity and its genetic basis. Based on the previous structural phenotype analysis that focused on the right insula, we still used the right insula as the seed point to get the correlation coefficient as the rs-FC value between the right insula and other brain regions. After GWAS analysis, we found

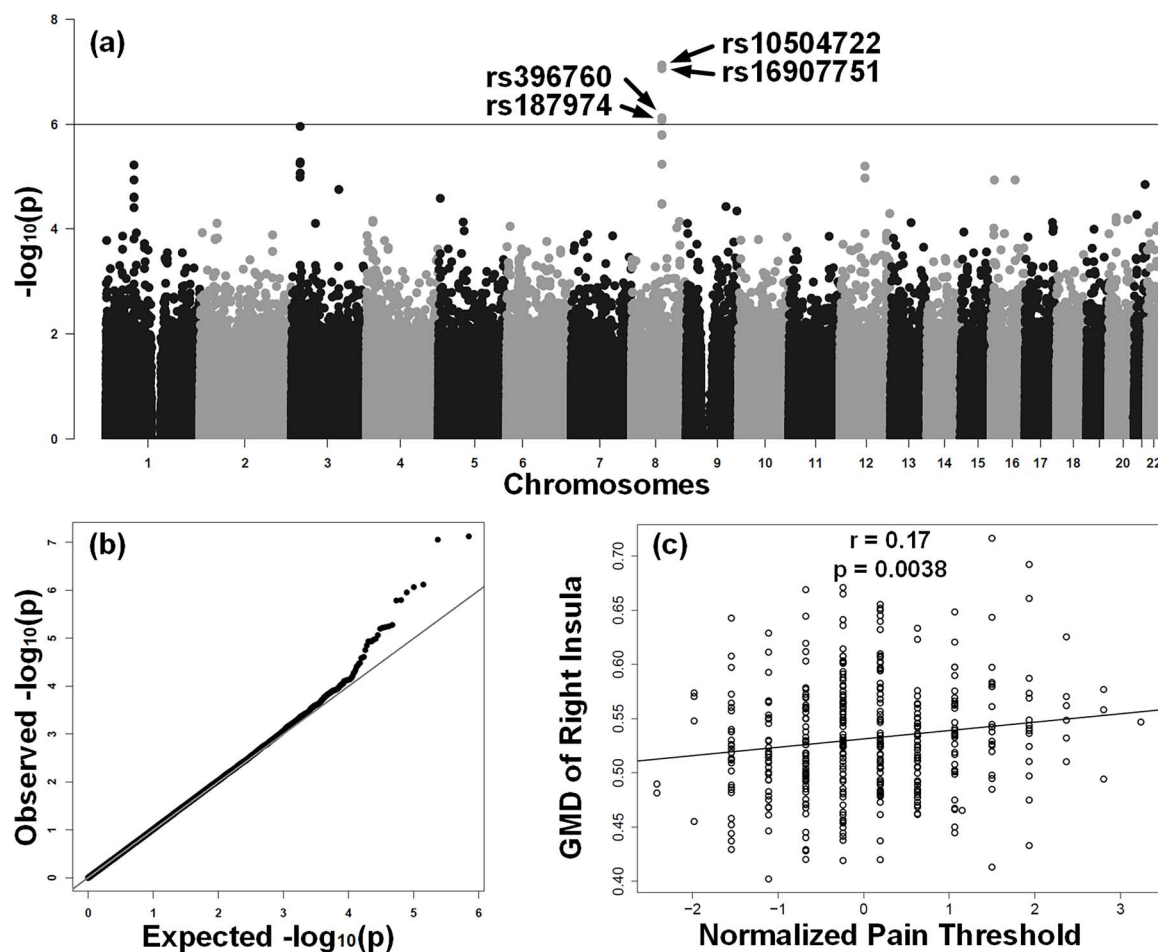


Fig. 5. Manhattan and Q-Q plots of GWAS of QT as the GMD of right insula, and correlation between GMD and normalized pain threshold. a) The horizontal line in the Manhattan plot indicates $P = 10^{-6}$. Four significant SNPs are marked with their names. b) The Q-Q plot of the right insula indicates the distribution of the observed P-values in this sample versus the expected P-values under the null hypothesis of no association. c) Correlation analysis revealed a positive correlation between GMD and normalized pain threshold ($r = 0.17$, $P = 0.0038$).

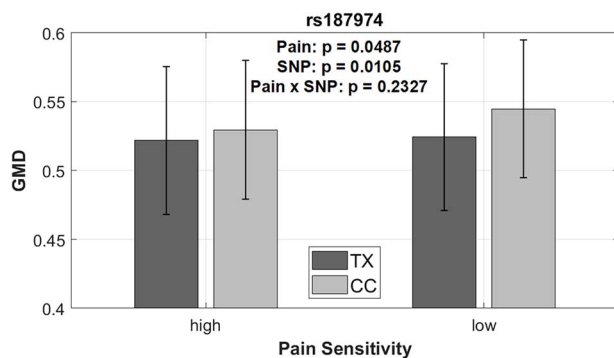


Fig. 6. Refined analysis of GMD of the right insula in relation to rs187974 and pain sensitivity group. A two-way ANOVA was performed to examine the effect of rs187974 and pain sensitivity on GMD of the right insula. The bars and error bars indicate the mean and SD of GMD.

that the rs-FC between the left putamen and right insula can be identified as a significant QT that correlated to SNPs rs2267447 and rs1081003. These 2 SNPs were extracted for subsequent statistical analysis. It is worth noting that both SNPs belong to the same gene named *CYP2D6* according to Single Nucleotide Polymorphism Database (<https://www.ncbi.nlm.nih.gov/snp/>). Interested readers may refer to Section 2 of the [Supplementary Document](#)

for the Manhattan plot of the GWAS of QT as the rs-FC between the right insula and left putamen.

We further conducted the statistical analysis using a two-way ANOVA to examine the effect of target SNPs and pain sensitivity groups on the rs-FC between the right insula and the left putamen. The results are shown in [Fig. 8](#). First, [Fig. 8a](#) and [b](#) indicates that participants with high pain sensitivity had significantly lower rs-FC values than low pain sensitivity for both target SNPs ($P = 0.0163$ and $P = 0.0246$). In [Fig. 8a](#), the main effect of the rs2267447 genotype across all participants was significant for the rs-FC value. More specifically, the rs-FC of the CC group was significantly lower than the TT group in the rs2267447 SNP ($P = 6.17 \times 10^{-5}$). In [Fig. 8b](#), we can also find that the rs-FC of the AA group was lower than the GG group in the rs1081003 SNP ($P = 6.08 \times 10^{-5}$). Both results suggest that participants with high pain sensitivity carrying C-allele of rs2267447 and A-allele rs1081003 may have a lower rs-FC value between the right insula and left putamen than rs-FC. We also conducted a correlation analysis between the normalized pain threshold and the rs-FC between the right insula and the left putamen. [Figure 8c](#) shows the relationship between pain sensitivity and rs-FC, and we can see that participants with higher normalized pain thresholds were found to have stronger FC between the right insula and left putamen.

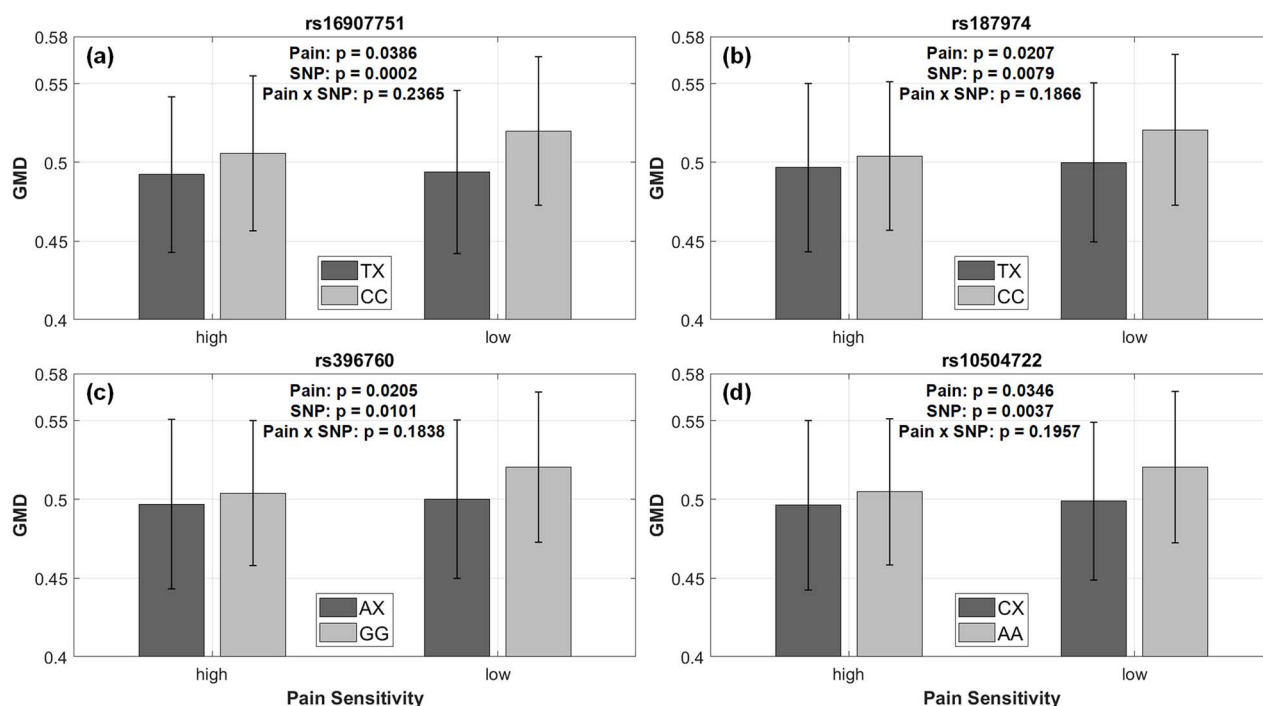


Fig. 7. Refined analysis of GMD of subregion dAI of right insula about corresponding SNPs and pain sensitivity group. Two-way ANOVAs were conducted to examine the effect of the 4 SNPs and pain sensitivity on GMD of the dAI. The main effect of the pain sensitivity group and a) rs16907751, b) rs187974, c) rs396760, and d) rs10504722. The P -values for the main effect of the pain sensitivity group and SNP are shown in the plot. The bars and error bars indicate the mean and SD of GMD.

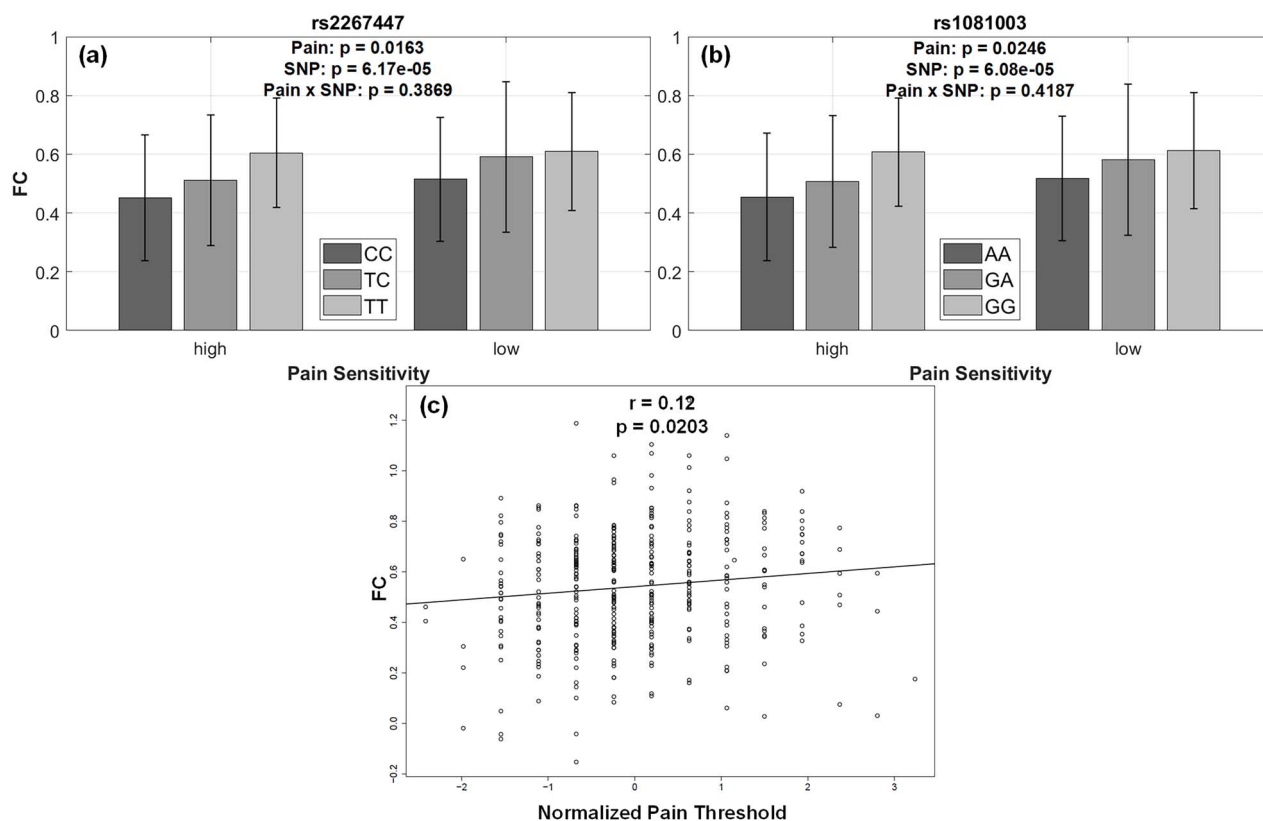


Fig. 8. Refined analysis of the rs-fMRI phenotype of right insula about corresponding SNPs and pain sensitivity group. Two-way ANOVAs were applied to examine the effect of the 2 SNPs and pain sensitivity on rs-FC between the right insula and the left putamen. The main effect of the pain sensitivity group and a) rs2267447 and b) rs1081003. Both SNPs belong to the gene CYP2D6. The bars and error bars indicate the mean and SD of rs-FC. c) Correlation analysis revealed a positive correlation between the normalized pain threshold and the rs-FC between right insula and left putamen ($r = 0.12$, $P = 0.0203$).

Furthermore, we have conducted a subregion analysis for the right insula. Similar to the GMD analysis, we divided the right insula into 3 subregions (PI, dAI, and vAI) to examine the rs-FC between each subregion and left putamen. Three GWAS analyses were performed for each rs-FC as QT. From the GWAS results, we found that 5 SNPs can be identified for the rs-FC between dAI and left putamen. Furthermore, a two-way ANOVA was performed to examine the effect of target SNP and pain sensitivity on the rs-FC phenotype of dAI and left putamen. As a result, all 5 SNPs show statistical significance for different genotype groups. However, no significant results can be found for the pain sensitivity group comparison. These results indicate that, although the rs-FC between right dAI and left putamen may vary among different genotype groups, it does not have a significant difference between high and low pain sensitivity groups. Interested readers may refer to Section 3 of the [Supplementary Document](#) for more information.

Discussion

Imaging phenotype definition

Pain sensitivity is a characteristic that affects the perceiving way of painful stimuli for different subjects. Though enormous progress has been made from previous studies, the reason why different individuals have different levels of pain sensitivity and what factors contribute to pain sensitivity individual differences are still open and challenging questions. Aiming these, we employed a whole-genome-whole-brain method in this paper to systematically identify the candidate QT loci associated with brain imaging phenotypes of pain sensitivity. The data analysis method can be summarized into 3 major steps: (i) preprocessing and QC, (ii) GWAS of brain structural phenotypes, and (iii) GWAS of brain rs-FC phenotypes. This approach is similar to other imaging genetics research ([Shen et al. 2010](#); [Stein et al. 2010](#)).

To investigate the brain structural and functional basis of pain sensitivity, the values of GMD from 170 ROIs were extracted. From the results shown in [Fig. 3](#), there are 98 brain regions that can be identified to be related to subjects' pain threshold. Most of these regions are located in the insula, basal ganglia, thalamus, and hippocampal regions. The strongest association area with pain threshold from our experiment is located in the insula cortex. These results are promising because a number of past studies have indicated that these regions play key roles in pain perception and modulation ([Barker 1988](#); [Chudler and Dong 1995](#); [Starr et al. 2009](#); [Borsook et al. 2010](#); [Lu et al. 2016](#)). For example, Lu has reported that the anterior portion of the insular cortex is critical for integrating and competing for multiple information to create awareness of the pain ([Lu et al. 2016](#)). Other studies have shown potential pathways via basal ganglia with respect to possible pain processing ([Barker 1988](#); [Chudler and Dong 1995](#)). Along with the findings from our results, these brain regions could be potential therapeutic targets for effective and individualized pain modulation, such as precision anesthesia and pain relief.

GWAS of structural brain phenotypes

In [Fig. 4](#), the heatmap and hierarchical clustering have been used to discover the SNP-by-ROI associations and the grouping results of different SNPs. One reason for using a heatmap is that it can give an intuitive representation of the associations between SNPs and target brain regions. These results also align well with the associated ROIs with pain threshold, as shown in [Fig. 3](#). The strongest associations include the right insula, right caudate nucleus, bilateral subgenual anterior cingulate cortices (ACC), and

bilateral amygdalae. If considering the number of associations as the criterion, the right insula shows the most significant result, as 4 SNP-by-ROI associations can be identified. Moreover, the left insula also shows identification with SNP rs16907751 at $P \leq 10^{-6}$ significance threshold. Though the other 3 SNPs cannot pass the given threshold test, the color map shows that they also have relatively stronger associations with the left insula than other ROIs.

From [Fig. 4](#), some structured patterns can be observed. On the one hand, in the imaging domain, the hierarchical clustering result on the x-axis shows the symmetrical property of bilateral brain hemispheres. Some of these ROIs, such as the insula, thalamus, ACC, and amygdala, belong to the conventional pain matrix, which is involved in the modulation of pain from top-down cognition and bottom-up tactile stimuli ([Lu et al. 2016](#); [Reddan and Wager 2018](#)). Meanwhile, the selection of 4 SNPs (rs10504722, rs16907751, rs396760, and rs187974) all associate with both left and right insula cortices. It is possible to identify the underlying brain connectivity between bilateral regions related to genetic variation. On the other hand, in the genomic domain, these SNPs are hierarchically clustered together to the eighth chromosome, as shown in the dendrogram on the y-axis in [Fig. 4](#). This is possibly due to the LD relationships for those SNPs. LD means that ≥ 2 loci at different loci are associated with each other ([Slatkin 2008](#)). More specifically, if random mating and no selection can be guaranteed for a large population, alleles should be randomly associated with each other. However, such an assumption cannot be met in reality for many reasons, such as genetic recombination, genetic drift, population structure, and genetic linkage. As a consequence, some alleles are found together more often than expected. Many studies about insula and pain have been conducted in recent years. For instance, a former VBM-based study found that the morphological changes of the right anterior insular cortex can affect the pain areas of patients with fibromyalgia ([Liu et al. 2022](#)). In ([Ferraro et al. 2021](#)), the authors implied that the dysregulation of anterior insula activity can be regarded as a possible functional biomarker of chronic pain from a meta-analysis. Based on prior research, our result is reasonable to identify multiple candidate SNPs associated with the insula.

Based on the heatmap and dendrogram results, we can further analyze potential genetic variation and morphological alterations. Since the right insula has the most SNP associations, we select this ROI as the target phenotype to examine its whole genome mapping in [Fig. 5](#). First, the 4 identified SNPs can be confirmed to be top markers that pass the $P \leq 10^{-6}$ significance threshold. Second, there are several other SNPs from the same chromosome, which also show a relatively stronger association with the right insula. This is may also because of the existing LD relationship but cannot pass the given significance threshold. Third, though the given significance threshold cannot be reached, several SNPs from chromosome 3 also show relatively stronger associations with the right insula. It is still worth conducting further validation research on these results.

To verify the identification of the right insula from our result, a correlation analysis was used to reveal a relationship between its GMD and the pain threshold. As shown in [Fig. 5c](#), one can see an increasing trend in the normalized pain threshold that has an increased density of the right insula. It implies a negative correlation between GMD and pain sensitivity, i.e. one with higher GMD of the insula may less sensitive to pain stimuli. Despite the arguable heterogeneity ([Ruscheweyh et al. 2018](#)), it is now generally accepted that pain perception has an association with GMD. For example, several independent studies have suggested

that chronic pain patients have a decrease in GM in common, and the locations overlapped in the insula, cingulate cortex, and dorsolateral prefrontal cortex (Apkarian et al. 2004; Schmidt-Wilcke et al. 2005; Draganski et al. 2006; Schmidt-Wilcke et al. 2006; Kuchinad et al. 2007; Rodriguez-Raecke et al. 2009). Meanwhile, the morphological changes of GM can be observed in phantom pain patients due to amputation and spinal cord injury (Draganski et al. 2006; Wrigley et al. 2009). One possible explanation is that cell atrophy and the decrease in cell size or blood volume can cause stronger pain syndromes (May 2008). Although there is no conclusive research, our result implies that the structural alteration of insula GMD may affect individual pain sensitivity.

A refined statistical analysis based on a chosen SNP-by-ROI association, i.e. the right insula and the SNP rs187974, was further performed. Figure 6 shows the result of a two-way ANOVA of GMD of the right insula in relation to rs187974 and pain sensitivity group, where 2 genotypes TT and TC together as TX due to the limited number of TT subjects. The first observation of Fig. 6 is that the TX group has a lower density of GM than the CC group for both high and low sensitivity subject groups with $P = 0.0105$. This suggests that the carrier of allele T at SNP rs187974 may lead to a lower GMD in the right insula than in CC subjects. Another observation from Fig. 6 is that the GMD in the high pain sensitivity group is slightly lower than the low sensitivity group, with a P -value of <0.05 . This statistical result verifies the finding in Fig. 5c that subjects with lower GMD may be more sensitive to pain. A detailed analysis was further conducted on the subregions of the insula, including the PI, dAI, and vAI. The result in Fig. 7 shows that the SNP rs187974 is only found to be associated with dAI. This could suggest that the GMD of dAI, but not the entire insula, may affect the individual pain perception. In line with our findings, Wiech et al. (2010) have demonstrated that the anterior insula encodes the perceived threat value of an upcoming laser stimulation before a stimulus encounter. Evidence was provided that the anterior insula structure plays a pivotal role in integrating information about the threat level of stimulation into the decision about pain. Many previous studies have also indicated that the anterior insula can anatomically integrate different types of information into sensory processing (Mesulam and Mufson 1982; Paulus and Stein 2006). Meanwhile, the functional connectivity of dAI has also been demonstrated to have variations for different pain conditions. For example, in Hong et al. (2014), by comparing FC of dAI in healthy subjects and patients with chronic abdominal pain conditions, differences can be identified for FC between bilateral dAI and dorsal medial prefrontal cortex for female subjects. Together with our results, these findings may imply that the anterior insula plays a key role in individual interoception that involves sensation monitoring. Further beyond the imaging dimension, our study also shows other possible bases for such findings from the genetic dimension.

From Fig. 7, statistical results show a significant association between the other 3 SNPs and dAI. The two-way ANOVA reveals that the GMD of dAI has statistically significant effects on both genotypes and pain sensitivity. However, there is not a statistically significant interaction between genotype and pain sensitivity from all these results. The SNPs rs10504722 and rs187974 belong to the gene ZBTB10 (Zinc Finger and BTB Domain Containing 10). The genes of the other 2 SNPs rs16907751 and rs396760 are limited known, and it is most likely that they also belong to this same gene due to the hierarchical clustering result from Fig. 4 and the LD relationship. Though indeed a literature search did not reveal any knowledge associating ZBTB10 and pain sensitivity,

additional analyses on such SNPs and genes appear possible for future study.

GWAS of functional brain phenotypes

There are 2 reasons for choosing rs-FC as functional brain phenotypes instead of task-related fMRI features. One is the prediction ability of the rs-FC network for individual pain sensitivity. In Spisak et al. (2020), it is identified and validated that the pattern in pain-free rs-FC networks are predictive of individual pain sensitivity. Several characteristics of the rs-fMRI signal are known to be associated with the neural response to nociception and the pain experience (Rogachov et al. 2016; Tu et al. 2019). Another reason is that rs-fMRI is more convenient and easier to obtain. Usually, painful stimulations are required for pain sensitivity for task-related fMRI studies. However, such actions and reliable behavioral pain reports may not be available for some patients. Using rs-fMRI allows us to assess the individual pain sensitivity without applying any painful stimulation during the fMRI experiment.

As we can see from Fig. 8, 2 SNPs rs2267447 and rs1081003 can be identified to be associated with the rs-FC between the right insula and left putamen. The main effect analysis of two-way ANOVA reveals that both genotype and individual pain sensitivity have a statistically significant effect on rs-FC strength. More specifically, the carrier of the C allele of rs2267447 and A allele of rs1081003 may lead to a lower rs-FC value. Meanwhile, the rs-FC between the right insula and left putamen in the high pain sensitivity group shows a lower value than in the low sensitivity group. Similar to the insula, the putamen is also considered to be part of the pain matrix. It is one of the major sites of cortical input into basal ganglia loops and is frequently activated during pain. Its role is often been assumed to be related to the processing of pain-related motor responses (Starr et al. 2011). From the aspect of brain structure, the putamen is part of the striatum together with the caudate nucleus. The region is rich in opioid receptors and nociceptive neurons that respond to varying levels of painful stimuli (Baumgartner et al. 2006). Accumulating evidence indicated that patients with many chronic pain syndromes, such as fibromyalgia, chronic vulvar pain, or chronic low back pain, show structural changes in striatal GM (Schmidt-Wilcke et al. 2006, 2007; Schweinhardt et al. 2008). Furthermore, studies of striatal activation striatal dopamine D2 receptor activity also indicated that this region is correlated with individual variability in pain and pain modulation (Hagelberg et al. 2004; Scott et al. 2006). The result of our study implies that the dysregulation of rs-FC between the right insula and left putamen may perform as a functional marker for pain sensitivity.

The identified 2 SNPs rs2267447 and rs1081003 both belong to the gene CYP2D6 (Cytochrome P450 Family 2 Subfamily D Member 6). CYP2D6 can be expressed in the brain, liver, and other extrahepatic tissues, where it metabolizes a range of centrally acting drugs and toxins (Miksys et al. 2002). It encodes the enzyme that is highly expressed in areas of the central nervous system. The cytochrome P450 family proteins are monooxygenases that catalyze many reactions involved in drug metabolism and synthesis (Bertilsson et al. 2002). The CYP2D6 gene is known to be highly polymorphic and is critical for opioid therapy to improve pain management in clinical scenarios (Smith et al. 2019). The instability of genes is believed to be responsible for the diversity of drug response (Leff and Yang 2015). Specifically, the polymorphisms of CYP2D6 may alter the pharmacokinetic mechanisms controlling the local availability of active analgesic molecules at their effector sites. As one of the analgesics on the World Health Organization

(WHO) essential drug list, codeine is metabolized by this gene to morphine (Susce et al. 2006). It has been demonstrated that codeine is under strong pharmacogenetic control by CYP2D6 and displays significant interethnic differences (Lotsch et al. 2009). Though it remains unknown how this gene could change the rs-FC between the insula and the putamen, the findings in this study suggest that the CYP2D6 gene could be a potential therapeutic target for pain modulation.

Limitations and future directions

Though the scale of our self-collected dataset can be considered as large in the field of pain sensitivity research, the number of included subjects is still relatively low for a brain-wide genome-wide study. This first limitation does not allow us a further validation study to divide the subjects into a test group and a validation group. Consequently, in the heatmap result, the significance threshold level at $P \leq 10^{-6}$ was reported. As a GWAS research, this operation is similar but slightly more relaxing than the Bonferroni correction method. More specifically, 353,421 SNPs were preserved after the QC procedures in this study. A strict Bonferroni correction should set the significant cutoff at $\alpha/n = 0.05/353,421 = 1.41 \times 10^{-7}$, where α is the significant threshold and n is the number of subjects. However, the Bonferroni correction tends to be overly conservative to catch some significant findings. Though in some GWAS previous studies (Zhang 2016; Chen et al. 2021), a P -value cutoff at 10^{-7} or 10^{-8} level was suggested to survive the multiple test corrections, here, we follow Shen et al. (2010) to use a less conservative P -value threshold to increase the true-positive SNP-QT associations discovery. Though a large dataset can usually lead to more robust results, our findings can also be supported by many other studies on the topic of pain. Future studies could incorporate additional participants in the GWAS research to examine the effects of SNP and target imaging phenotypes. Second, this study did not consider the gene expression of the identified SNPs or the gene-gene interactions. These are very important topics for understanding the pathway of pain modulation and the mechanisms of the difference in individual pain sensitivity. Future analysis should employ the gene expression dataset, such as AHBA, for more comprehensive research to reveal more important factors in pain sensitivity. Third, the AAL3 atlas was used in this study to create ROIs. In the future, a more detailed atlas or probabilistic atlas can be employed to enhance the resolution of imaging phenotype.

Conclusion

To conclude, we carried out a whole-genome-whole-brain pain sensitivity study in a self-collected dataset with 432 normal participants. We provide evidence that the right insula and its subregion dAI are significantly associated with individual laser pain sensitivity and can play an important role in pain perception. Moreover, GWAS and a series of refined analyses demonstrated potential key loci and their associated brain structural and functional phenotypes. Some of the encouraging results and findings are in line well with the existing literature. It indicates that the existence of genetic variants may contribute to individual differences in pain sensitivity. This study is one of the few to investigate the brain structural, functional, and genetic basis of pain sensitivity using an imaging genetic approach. Overall, our results provided several potential QT loci associated with the structural and functional phenotypes of pain sensitivity. These findings may provide more insight into the mechanisms of underlying pain sensitivity in humans and may

also inform the future development of clinical precision pain modulation.

Authors' contributions

Li Zhang (Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Visualization, Writing—original draft, Writing—review & editing), Yiwen Pan (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing—original draft, Writing—review & editing), Gan Huang (Methodology, Resources, Validation), Zhen Liang (Resources, Validation, Writing—review & editing), Linling Li (Resources, Validation, Writing—review & editing), Min Zhang (Validation, Writing—review & editing), and Zhiguo Zhang (Conceptualization, Data curation, Investigation, Methodology, Project administration, Supervision, Validation, Writing—original draft, Writing—review & editing)

Supplementary material

Supplementary material is available at *Cerebral Cortex* online.

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